

Will AI Replace Doctors?

The planning guide — the full scores by track, the six factors behind them, the honest trade-offs, and what to ask before you commit.

2.1

AI EXPOSURE · SURGERY

where 10 is most at risk

For anyone still choosing — read it straight through.



Before you read this

You're weighing a medical degree — or a path into the profession — and you want to know what that choice actually looks like in an AI economy. That's the whole job of this guide.

It's written to you directly, whether you are the one choosing or a parent thinking it through alongside them. Where it says "you," a parent can read "them"; it's the same decision from either side.

What this is. A facts document, not a recommendation. It sets out what's known, where the evidence is contested, and where we might be wrong — including the parts that don't support our own conclusions — and then how to choose well given all of it.

If you're the parent, a note on how to use it. Don't lead with the scores. Open a conversation with a risk number and one of two things happens: they shut down, or they get defensive about a choice they're already leaning toward. Read it yourself first, then share it as information you both react to, not a verdict you've reached. They're choosing, not you — the job is to widen what's been considered, not to narrow it.

And one thing specific to medicine. It is a longer commitment than almost anything else you can choose — ten to fifteen years from starting a degree to independent practice. If the certainty is already there, that certainty is an asset; don't unpick it. Use this to understand what the commitment involves, and to choose a specialty knowingly when the time comes — because for the first time, that choice carries an AI dimension it didn't a few years ago.

The short answer

Yes — medicine remains one of the safest paths available, and patient-facing practice holds the strongest protection stack in this entire index. But for the first time, choice of specialty carries an AI dimension. A student entering medical school today will choose a specialty in five or six years, and the specialties are no longer moving together.

Surgery scores 2.1 — the lowest score of any career we measure. Radiology scores 4.9 and is the fastest-moving protected career we track. Same degree, same training, a 2.8-point gap.

The scores

SPECIALTY	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Surgery	1.8	1.9	2.1	+0.3	Very low
Emergency medicine	2.0	2.2	2.5	+0.5	Very low
Primary care / family medicine	2.4	2.6	2.9	+0.5	Low
Anesthesiology	2.7	2.9	3.1	+0.4	Low
Psychiatry	3.1	3.4	3.7	+0.6	Low
Pathology	3.8	4.3	4.7	+0.9	Moderate
Radiology	3.9	4.4	4.9	+1.0	Moderate

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, bedside nursing scores 2.8, and entry-level software development scores 8.1.

Read the table this way: every specialty in medicine sits below the index median, so the profession as a whole is exceptionally well protected. But the spread has widened from 2.1 points in 2023 to 2.8 today, and it widens every year we measure it. Medicine is not becoming riskier. It is becoming less uniform.

Why medicine scores low — the six factors

How much of this job can AI already do?		6.0
How hard will it be to land that first job?		1.5
Does it have to be done in person, with your hands?		8.0
Does someone need a human they can trust and hold responsible?		9.5
Does the law require a licensed human?		9.5
How often does the job hit genuinely new, high-stakes situations?		8.5

EXPOSURE RAISES THE SCORE PROTECTION LOWERS IT

We score every profession in this index against the same six questions, with the same weights. Here's how primary care answers them — the reference point for patient-facing medicine.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

A substantial share — and it is the layer around the medicine, not the medicine.

Clinical documentation, letters, coding, record summarization, guideline and drug-interaction lookup, and increasingly triage and risk stratification. Ambient systems that draft the consultation note are already in routine use, and most physicians regard them as a relief rather than a threat.

What is not automating is the physical examination, the decision made under genuine uncertainty, and the conversation with a family when the news is bad.

This is the one factor where specialties genuinely diverge, and the radiology question further on covers that in detail.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

The most protected on-ramp we measure anywhere in this index.

Across almost every other profession, AI is absorbing the junior tasks first — the grunt work that entry-level roles were built from, and that people historically used to build judgment. The professions survive; the paths into them collapse.

Medicine is structurally immune. Residency is mandated, funded, structured and legally required — a doctor cannot exist without supervised years. Whatever else changes about the profession, its apprenticeship is written into the system in a way no employer can quietly erode.

The way through, here, isn't about surviving the on-ramp — medicine's is the one entry route in this index that holds. It's about what you steer toward once you're on it. The divergence is entirely in specialty choice: the patient-facing and procedural specialties (surgery 2.1, emergency medicine 2.5, primary care 2.9) hold their protection, while the diagnostic, screen-based ones (pathology 4.7, radiology 4.9) carry more exposure. That's a set of concrete choices, not a hope — and the second half of this guide is about making them deliberately.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

For most specialties, substantially yes.

Examining a patient. Operating. Delivering a baby. Resuscitating someone. These are irreducibly physical, and robotics is nowhere near them.

This is also where the internal spread begins: a surgeon rates near the top of this factor and a radiologist near the bottom, and that single difference does more work than any other in

separating a 2.1 from a 4.9.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Absolutely, and in a form that is unusually hard to delegate.

Nobody wants an algorithm to tell them they have cancer. And when a clinical decision goes wrong, a licensed human carries the responsibility — legally, professionally and morally. That accountability attaches to a person, not to a system, and no advance in capability transfers it.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

Comprehensively, and more thoroughly than almost any other profession.

Licensure, registration, prescribing authority, scope-of-practice rules and malpractice liability all mandate a qualified human. This protection moves at the speed of legislation and public trust, not at the speed of technology — which is why we rank it the most durable protection in the entire framework.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Constantly.

Patients present atypically as a rule. They arrive with several conditions interacting, deteriorate unpredictably, and don't match the textbook. This is precisely where AI remains weakest — genuine novelty, where the pattern doesn't fit and being wrong is expensive.

What the trend shows

We retrospectively scored medicine for 2023 and 2025 using this same methodology. The interesting finding is not the movement — it's the divergence.

Patient-facing medicine: 2.4 → 2.6 → 2.9. Up 0.5 points in three years, and nearly all of it administrative automation. The physical, trust, licensing and judgment scores have not moved at all.

Diagnostic specialties: 3.9 → 4.4 → 4.9. Up a full point — the fastest-moving protected career we score anywhere.

For comparison over the same period, an entry-level software developer moved 1.8 points. Medicine is drifting; software is sprinting. But within medicine, radiology is moving at twice the rate of the specialties around it.

The gap between the safest and most exposed specialty has grown from 2.1 points to 2.8. Three years ago you could reasonably say "medicine is safe" and stop there. That is no longer a complete answer.

How durable is the protection?

Every "AI can't do X" boundary has proven temporary. It couldn't do nuance, then it could. Couldn't do images, then it could. So the useful question isn't whether medicine's protections hold today — they clearly do — but how long each is likely to last.

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Regulatory / licensing	Strongest available	Most durable	Moves at the speed of law. No capability advance repeals a licensing requirement or malpractice liability.
Human trust / accountability	Very strong	Durable	Socially rooted. People want a human present at frightening moments, and someone must answer for the outcome.
Judgment under novelty	Strong	Eroding	Diagnostic support and risk stratification are advancing — currently assisting, not deciding.
Physical / embodied	Strong for most specialties	Most fragile long-term	The protection physical AI directly targets. Surgical robotics is assistive today; the horizon is longer than a career.

The honest read: you'll practice for roughly forty years. Over that span, the two protections that will still be standing are the ones rooted in law and society rather than in what the technology can do. Physical protection is the strongest shield today and the one most likely to weaken within a career.

Note also which protections are *specialty-dependent*. Licensing and accountability protect every doctor equally. Physical presence and novelty do not — and that is precisely why the specialties are pulling apart.

How the role will actually change

Not "will medicine exist" — it will. The useful question is what the job looks like in ten years.

WAS THE JOB	IS BECOMING THE JOB
Writing the note after the consultation	Ambient systems draft it; the physician verifies and corrects
Recalling the guideline	Decision support surfaces it; the physician judges whether it fits this patient
Reading the scan	Supervising the AI read, adjudicating the ambiguous cases, owning the decision
Documentation consuming the evening	More time with patients — the part most doctors came for
Triage by experience	Risk stratification flags the deteriorating patient earlier

The pattern across this index is that AI absorbs the routine, repeatable layer first. In creative and knowledge work that layer was often the craft itself. In medicine it is the paperwork, which is the leading cause of physician burnout. Automating it returns time to patients rather than removing the practice.

Two honest caveats. Verification of AI-generated notes and reads becomes a real clinical responsibility — a wrong record is a patient-safety issue, not an administrative one. And efficiency gains can be captured as higher patient loads rather than returned as time, which is a funding decision rather than a technological one.

Human strengths that will matter most

Ranked by how well each resists the coming decade, specific to medicine:

1. **Physical examination and clinical intuition** — noticing the patient who looks wrong before the numbers say so. The hardest thing to replicate and the most valuable thing a doctor builds.
2. **Judgment under genuine uncertainty** — the atypical presentation, the interacting conditions, the case with thin precedent and high cost of error.
3. **Accountability** — being the human who owns the decision. Durable, legally rooted, and the reason a qualified person stays in the loop.
4. **Communication at the hardest moments** — breaking bad news, negotiating a treatment plan, being trusted by a frightened family.
5. **Verification and discernment** —

knowing when an AI-generated note, read or recommendation is subtly wrong. A genuinely new core skill, and one barely taught anywhere yet.

Notably absent: speed of recall and breadth of memorized knowledge. These were the traditional markers of a strong medical student and are the fastest-depreciating.

And underneath all five, two traits the score can't grade: curiosity and adaptability. The ambient documentation, decision support and diagnostic tools will turn over many times across a forty-year career. The doctors who compound are the ones who stay genuinely curious about the medicine and re-tool fastest, rather than defending the one way of working they trained on. In a profession where the automatable layer is the paperwork rather than the practice, that flexibility is itself a form of protection — and increasingly it's what the informatics and leadership paths are looking for: the trajectory, not the current tool.

Other things to consider about this job

Everything above measures one thing: how exposed medicine is to AI. That's useful, and it's deliberately narrow. It says nothing about whether the work is rewarding, whether the training is survivable, or who this suits.

All of that matters more to a real decision than the score does. So here it is, in four parts.

What's genuinely good about it

WHAT DOCTORS THEMSELVES SAY

The satisfaction picture in medicine is genuinely two-sided, and both sides are real.

Physician job satisfaction sits at 77% and has risen every year since 2022, according to the AMA's national survey. Satisfaction is highest among physicians in their first five years out of training and those twenty years or more in — the difficult stretch is the middle.

What they report as rewarding is remarkably consistent:

- **Being trusted at the most consequential moments of someone's life** — the relationship, not the transaction
- **Intellectual depth** — medicine remains one of very few careers where the learning genuinely never stops
- **Mastery of something difficult** — competence in a demanding craft, earned over years

- **Breadth of choice** — few qualifications open onto so many genuinely different working lives
- **Autonomy at senior level**, and in many systems the option of independent practice

THE PRACTICAL CASE

- **Exceptional job security.** Physician shortages are projected across most developed health systems, and the training pipeline is constrained by funding rather than demand.
 - **A portable qualification**, though re-licensing across countries is more onerous than for nursing.
 - **Strong earnings**, with wide variation by specialty and system.
 - **Genuine specialty choice** — someone who wants procedures, or continuity, or acute pressure, or research, can find a version of this career that fits.
-

What's genuinely hard about it

In medicine, automation is not the biggest risk to this career. The length and cost of getting there, and burning out once you have, are the real ones.

THE LENGTH

This is the single most under-appreciated fact about medicine, and it is rarely held concretely in mind.

From starting a medical degree to independent practice is typically **ten to fifteen years** depending on specialty and country. That period includes sustained examination pressure, repeated relocation for training posts in most systems, and years of working under supervision at a stage when peers in other careers are already established.

THE COST

In systems where medical education is paid for privately, the debt is substantial and it compounds during years of relatively low training-grade income. This is a genuine constraint on who can realistically enter the profession, and it should be discussed openly rather than assumed away.

BURNOUT AND MORAL INJURY

Physician burnout is high — but it has been **falling for three straight years**, and this is one place the popular narrative is out of date. The AMA's 2025 survey of nearly 19,000 physicians

across 106 health systems found 41.9% reporting at least one burnout symptom, down from 43.2% in 2024 and 48.2% in 2023.

It remains meaningfully above the general working population, and it is very unevenly spread: emergency medicine, urological surgery and hematology/oncology all sit near or above 49%. Around 31% of physicians report intending to leave their organization within two years.

The most-cited drivers are administrative burden, loss of autonomy, and the specific distress of being unable to provide the care you believe a patient needs because of time, staffing or system constraints.

That last one has a name — **moral injury** — and it is distinct from being tired. It is what makes people leave a profession they still believe in.

EARLY-CAREER HOURS

Training-grade hours remain demanding in most systems, with night and weekend work through the years when many people are also forming families.

Who this work suits — and who it doesn't

Fit predicts staying far better than aptitude does. Medicine tends to suit people who:

- **Can hold uncertainty without freezing.** Much of clinical practice is deciding well with incomplete information. Some people find that energizing; others find it corrosive.
- **Get calmer when things go wrong.** A genuine and identifiable trait, hard to teach, and disproportionately valuable.
- **Are genuinely interested in people, not just in biology.** The science is the entry ticket; the practice is relational.
- **Can sustain a long horizon.** The rewards arrive late. Someone who needs visible progress every year will find the training years hard.
- **Tolerate being wrong in public.** Medicine involves error, review, and being corrected by colleagues, throughout a career.

Equally honest about who finds it hard:

- People who need **control over their own schedule** will struggle for a decade or more.
- People who need **certainty and closure** find the ambiguity difficult.

- People drawn primarily by **status or parental expectation** are heavily over-represented among those who leave.

Squeamishness, for what it's worth, is almost never the barrier people expect — nearly everyone acclimatizes faster than they think.

What else is moving this market

The AI score measures task exposure. It cannot see supply, demand, funding or the politics of health systems — and those often matter more.

Training capacity is the constraint, not demand. Medical school places and residency positions are capped by funding rather than by applicant quality or patient need. That protects the profession's scarcity, and it also means competition for entry stays fierce regardless of how many doctors a country needs.

Satisfaction varies more between specialties than the profession-level numbers suggest. Psychiatry reports 83% job satisfaction and obstetrics 81.2%, while hospital-based specialties — emergency medicine, radiology, anesthesiology — sit lowest at 74.8% and perform below benchmark on most wellbeing measures. That is a reminder that the setting matters as much as the specialty, and neither is visible in an AI score.

Shortages are structural in most systems, particularly in primary care, psychiatry and several procedural specialties. This is a profession where the labor market has been tight for decades and is projected to stay tight.

Working conditions are the volatile part. Pay, hours, administrative load and autonomy move with health-system funding and politics rather than with technology. Understand that the technology outlook for medicine is stable and the conditions outlook is not.

And one specialty-specific note. Radiology deserves flagging here because its labor market and our score point in opposite directions — see the radiology question, below.

The radiology question — where our own score is contested

We flag this because it is the strongest evidence against any number in this index, and it concerns the specialty most students ask about.

The case for our 4.9. Image interpretation is pattern recognition on structured visual data — the single thing AI does best. Around three-quarters of all FDA-authorized AI-enabled medical devices are in radiology, because that is where the technology works. The score has moved a full point in three years, faster than any other protected career we track.

The case against it. The labor market has done the opposite of what the narrative predicts. Roughly half of US radiologist job openings went unfilled in recent years. Imaging volume is growing 3–4% annually against a residency pipeline capped by federal training funding — a supply constraint with nothing to do with AI. Diagnostic radiology filled at about 97% in the 2025 Match, and interventional at essentially 100%. And there is little evidence so far that AI has delivered substantial productivity gains to the specialty.

Radiology has been the poster child for imminent AI displacement for roughly a decade, and the shortage has deepened throughout.

Where we land. We hold the score, and we publish the disagreement alongside it. Exposure to automation is not the same as exposure to unemployment, and in radiology the gap between those two is wider than anywhere else we measure. Bending a number to match an observed outcome would make every trend in this index meaningless.

One subtlety worth carrying. Radiology applications have declined from their 2023 peak, and reporting attributes part of that to student uncertainty about AI. So the fear is already shaping behavior even where the labor market isn't — which is arguably a reason a document like this should exist.

What we're watching: whether radiologist vacancy rates start falling, and whether AI-assisted reporting produces measurable throughput gains. Either would move the score, and we'll say so.

The AI-native advantage — what to actually do about it

This is the section you can act on. Everything else describes a landscape; this is about position within it.

The situation, factually: AI is already in clinical settings. As of early 2025 the FDA had authorized 882 AI-enabled medical devices, and the large majority of US hospitals use some form of AI-enabled clinical decision support. Meanwhile most practicing physicians are encountering these tools without any formal training in them.

That gap is the entire opportunity. The technology is deployed. The people who know how to use it well are not.

FIRST, A WORD ABOUT HOW SCHOOLS FRAME THIS

Most of what you hear about AI at school is about cheating. Don't use it for your essay. We can detect it.

Those rules exist for good reason, and they're not wrong. But they leave students with a badly incomplete picture, because they treat AI as a temptation to be resisted rather than a technology to be understood. Those are different things, and only one of them is relevant to a career.

- **Using AI to avoid the work** — writing your essay, generating your answer, skipping the reasoning — leaves you weaker. In medicine this eventually becomes a patient-safety problem rather than a grades problem, because the verification skill below only works on clinical judgment you actually built yourself.
- **Understanding AI** — what it does well, where it fails, how it behaves in a clinical context — is a professional competency. It is not cheating and never was.

WHAT AN AI-NATIVE DOCTOR ACTUALLY LOOKS LIKE

Not someone who has memorized tools. Someone who can do five things:

- 1. Verify.** Knowing when a generated note, read or recommendation doesn't match the patient in front of them. This is a clinical skill, not a technical one — it requires knowing the patient well enough to catch a plausible-sounding error.
- 2. Escalate.** Knowing what to do when the output is wrong: who to tell, how to document it, how to override safely. Institutions are still writing these policies.
- 3. Explain.** Being able to tell a patient, in plain language, what a system contributed and why a human remains accountable. Trust is medicine's deepest protection; someone who can articulate it under scrutiny defends it.
- 4. Recognize the limits.** Understanding at a working level why models produce confident errors, and how algorithmic bias shows up differently across patient populations. This is the ethics dimension and it is increasingly examined.
- 5. Contribute.** Being the person who can say whether a system works in practice. Physicians who can speak to implementation get pulled onto committees, into informatics, and eventually into leadership.

CONCRETE PREPARATION, BY STAGE

Before medical school:

- Use the general tools enough to develop instinct for where they're confidently wrong. That instinct is transferable and free to build.
- Follow how AI is landing in healthcare specifically — the medical press covers it continuously.
- Ask medical schools how AI appears in the curriculum. Their answers will differ enormously.

During the degree:

- Treat clinical informatics as a real subject rather than a box to tick.
- On placement, notice which systems each department actually uses and where senior clinicians work around them. Workarounds show you where a system fails.
- Practice the verification habit deliberately: when a tool suggests something, ask what would make it wrong before accepting it.

Choosing a specialty:

- Weigh the AI dimension, but don't let it dominate. The scores here span 2.1 to 4.9 — a real spread, but every one of them is below the index median.
- If two specialties genuinely appeal equally, the more physical and more patient-facing one carries more durable protection. If one appeals much more, choose that one.

WHY THIS IS AN ADVANTAGE RATHER THAN A BURDEN

Every generation of doctors has absorbed a technology shift. What's different here is timing: this cohort arrives while competency frameworks are still being written and while most practicing physicians are learning on the job.

A newly qualified doctor who is genuinely fluent stands out immediately against colleagues with decades more clinical experience — not on clinical skill, which takes years, but on a dimension where seniority confers no advantage.

The realistic version: this won't make anyone a better doctor than a great experienced one. It will make them useful faster, more visible to the people who staff projects and committees, and better positioned for the informatics and leadership paths covered under where a medical degree can take you later.

The routes in — they're not all the same

ROUTE	TYPICAL LENGTH	NOTES
Direct-entry medical degree	5–6 years	The standard route from school. Entry is the most competitive stage.
Graduate-entry medicine	4 years post-degree	For someone who studies something else first. Relevant if you're uncertain now.
International medical degree	5–6 years	Widely used where domestic places are scarce. Verify licensing recognition in the country where you intend to practice before committing — this is where families most often get caught out.
Physician associate / assistant	2–3 years post-degree	A genuinely different career rather than a lesser one: clinical work, shorter training, less autonomy and lower ceiling. Worth knowing exists.
Specialty training	+3–8 years post-degree	Where the specialty choice is actually made, and where the scores in this guide start to matter.

Two things worth checking locally: how competitive entry currently is where you are, and whether the intended route is recognized for licensing where you want to practice. The second one is the expensive mistake.

Where a medical degree can take you later

One of the strongest arguments for medicine is that it doesn't strand anyone. If the clinical path stops suiting you, the qualification remains valuable.

- **Specialty practice** — the main path, with enormous variation in rhythm, pressure and lifestyle
- **Academic medicine and research** — clinical trials, translational research, teaching
- **Clinical informatics** — the bridge between practice and health IT. The people who understand both the ward and the software are exactly who is needed to implement and supervise the systems described above.
- **Public health and policy** — population health, epidemiology, government and NGO work

- **Medical leadership** — clinical directorship, medical directorship, health-system management
- **Industry** — pharmaceutical medicine, medical devices, regulatory affairs
- **Medico-legal, occupational health, and expert practice**

The caveat that runs through this whole index applies here too: the further a role sits from the patient, the more its exposure climbs. Informatics and industry roles are genuinely good careers — and they score in the moderate band rather than the low one. That's a reason to move knowingly, not a reason not to.

What to look for in a medical school

The degree is protected. Not every program is equally good. Score any against these five.

- 1. Early and sustained clinical contact.** How soon do students see patients, and how often? Programs that front-load two years of lecture theater before meaningful clinical exposure are training for an older version of the job.
 - 2. Is AI treated as part of clinical practice?** Students should be learning to work with ambient documentation, decision support and diagnostic tools — including **where they fail**. A school that ignores them is preparing students for a job that no longer exists; one that teaches them uncritically is worse.
 - 3. Does it build reasoning, or recall?** Look for case-based learning, diagnostic reasoning taught explicitly, and assessment that tests judgment rather than memorized fact. Recall is the automatable part.
 - 4. Outcomes that mean something.** Not just pass rates — what proportion match into their preferred specialty, and where do graduates end up practicing?
 - 5. Support and attrition.** Medicine has real rates of burnout and withdrawal during training. Ask what support exists and how many students don't finish.
-

Questions to ask a medical school

Ask these directly. The quality of the hesitation tells you as much as the answer.

On clinical exposure:

- In which year do students first have sustained patient contact?
- How many weeks of clinical placement across the whole degree, and in what range of settings?
- How much of the clinical teaching is delivered by practicing clinicians?

On AI and technology:

- How does the curriculum prepare students to work with clinical AI tools?
- Where do students learn to identify when an AI-generated note or read is wrong?
- Has the curriculum changed in the last two years in response to these tools? What specifically changed?

On outcomes:

- What proportion of graduates match into their first-choice specialty?
- Where are your graduates practicing five years out?
- What's your attrition rate, and what are the most common reasons students leave?

On support:

- What mental health and wellbeing support exists, and how many students use it?
- What happens to a student who struggles in third year?

Red flags: evasiveness on attrition; "we don't use AI tools here"; late or minimal clinical exposure; a curriculum unchanged since 2022; outcome data quoted only as aggregate pass rates.

The career-hunting checklist

Choosing well is step one. This is what turns a good choice into a good outcome once you're in — worth reading now, because the best time to pick a program is knowing exactly what you'll need to do with it. The first list is for the degree and training years; the second is for the Match and job hunt that follow.

During the degree / training — build what can't be automated

- **Choose a specialty with durable protection, and choose it knowingly.** The patient-facing and procedural specialties hold their protection best — surgery (2.1), emergency medicine (2.5), primary care (2.9), anesthesiology (3.1), psychiatry (3.7) — while the diagnostic, screen-based specialties carry more exposure — pathology (4.7), radiology (4.9). Every one sits below the index median, so this is choosing between good and better, not safe and unsafe. If two specialties appeal equally, the more physical and more patient-facing one carries the more durable protection.
- **Build the durable clinical skills deliberately.** Physical examination and clinical intuition, judgment under genuine uncertainty, and communication at the hardest moments are the least automatable things a doctor builds — and the ones that separate the protected work from the automatable layer around it. Speed of recall and memorized breadth are the fastest-depreciating; don't mistake them for the skill.
- **Treat clinical informatics as a real subject.** On placement, notice which systems each department actually uses and where senior clinicians work around them — the workarounds show you where a system fails.
- **Practise verification as a habit.** When a tool suggests something — an ambient note, a decision-support prompt, an AI read — ask what would make it wrong before accepting it. This is a clinical skill, not a technical one, and it's barely taught anywhere yet.
- **Feed curiosity and adaptability on purpose.** The ambient documentation, decision support and diagnostic tools will turn over many times across a forty-year career; the doctors who compound are the ones who stay curious and re-tool fastest.
- **Keep the frame wide.** Understand the length before committing — ten to fifteen years to independent practice — and know that adjacent paths (nursing, physiotherapy, paramedicine, dentistry, physician associate, biomedical research) overlap with medicine but differ enormously in training length, cost and daily rhythm.

During the Match and job hunt — aim, don't just take what's open

- **Aim at the durable specialties on purpose.** Weight the AI dimension in the specialty decision, but don't let it dominate — the spread runs 2.1 to 4.9 and every point of it is below the index median. Where the human is in the room, the protection concentrates; where the work is reading patterns off a screen, it thins.
- **Read the specialty, not just the opening.** The protections rooted in physical presence, trust and accountability attach to the doctor at the bedside, in the theater, in the resus room. A post closer to the patient and to the decision you own is closer to the most durable protection medicine has.

- **Don't grab an exposed-lane role just because it's on offer.** A role being available doesn't make it safe. Roughly half of US radiologist openings went unfilled in recent years and diagnostic radiology filled at about 97% in the 2025 Match — the posts are plentiful, and the specialty still carries the highest AI exposure in medicine at 4.9. Exposure to automation is not the same as exposure to unemployment, so this isn't a reason to avoid it — only a reason not to read an open door as a settled answer on the automation dimension.
- **Move away from the patient knowingly, not by drift.** Clinical informatics, industry and medico-legal work are genuinely good careers — and they score in the moderate band rather than the low one. The further a role sits from the patient, the more its exposure climbs.
- **Ask where AI is heading in the role.** On placement or interview, whether a department is building the job around ambient documentation, decision support and real verification — or waiting to be forced to — tells you if the seat grows or is quietly being hollowed out.
- **Weigh the trajectory over the immediate.** A training post that builds the durable clinical, procedural and verification skills beats one that leaves you doing the automatable layer, however convenient it looks now.

Go deeper — further reading

The evidence:

- **How AI Will Shape the Future of Health Care in 2026** — SullivanCotter. The clearest statement of the pattern this guide describes: AI produces first drafts while clinicians concentrate on validation, interpretation and decision-making. Also covers why radiology has shown earlier proof points than other specialties.
- **Radiology in 2026: The Workforce Crisis Meets the AI Revolution** — Becker's. The two forces acting on diagnostic specialties simultaneously — a genuine shortage against reimbursement compression and accelerating AI adoption. Important because it shows the pressures are not purely technological.
- **AI solutions to the radiology workforce shortage** — npj Health Systems (Nature). Peer-reviewed framing of AI in radiology as a response to a supply shortage rather than a cause of displacement.
- **Physician burnout rate continues to decline, falling to nearly 42%** — American Medical Association. The primary source for the wellbeing figures above: nearly 19,000 physicians across 38 states and 106 health systems. Worth reading because the direction of travel

contradicts most popular coverage — burnout has fallen three years running while satisfaction has risen.

- **Physician burnout declines overall, but gaps persist across specialties** — Medical Economics. The specialty-level breakdown, and the finding that hospital-based specialties underperform on most wellbeing measures.

The counterargument — read this one:

- **Match Day 2026: Radiology programs offer more positions than ever** — Radiology Business.

This is the strongest evidence against any score in our index, and it targets our 4.9 directly. Roughly half of US radiologist openings went unfilled in recent years. Imaging volume grows 3–4% annually against a residency pipeline capped by federal funding. Diagnostic radiology filled at about 97% in the 2025 Match; interventional at essentially 100%.

Where we land: we hold that diagnostic work is more exposed than patient-facing work — image reading is what AI does best, and the concentration of FDA authorizations shows where the technology is. But exposure to automation is not exposure to unemployment, and in radiology the gap is wider than anywhere else we score. We publish the number with the disagreement stated rather than adjusting it to fit, and we will report it plainly if vacancy rates start to fall.

Bottom line

Medicine holds the strongest protection stack in this index — comprehensive licensure, deep trust, physical presence in most specialties, legal accountability, and constant novel judgment. Its entry route is the most protected we measure anywhere. That's the good news for someone still choosing: the risk question is largely settled before you begin.

The real questions are the ones families have always faced: ten to fifteen years to independence, the cost, the hours, and whether the vocation is genuine.

What this index adds is one thing: specialty choice now carries an AI dimension it didn't carry five years ago. The protection concentrates where the human is in the room, and thins where the work is reading patterns off a screen. That divergence is widening every year we measure it.

So the useful advice isn't "do medicine" or "avoid radiology." It's: do medicine if the vocation is real, understand the length before committing, and choose a specialty knowing that the specialties are no longer moving together.

And it's worth restating what the risk analysis can't measure. Doctors report among the highest sense of meaning of any profession measured. Being trusted at the most consequential moments of people's lives, mastering something genuinely difficult, and a career where the learning never stops — very few paths offer that combination at all. The hard parts above are real and shouldn't be minimized. So is that.

Questions worth sitting with

Whether you're the one choosing or a parent thinking it through together, these are the conversations that lead to a better decision. Read "you" as whoever the choice is about.

About the work itself

1. What made medicine appealing in the first place? Push past "helping people" to something specific — a moment, a person, a situation. 2. When you picture the job, what does a Tuesday look like? If a parent is reading too, compare pictures — they're often very different. 3. Do you want continuity with patients over years, or acute problems solved and moved on from? Those point to very different specialties. 4. How do you feel about being wrong in front of colleagues — regularly, throughout a career?

Reacting to the findings

5. Surgery scores 2.1 and radiology 4.9. Does that gap change how you think about the specialties you're drawn to — or does it feel too far away to matter yet? 6. This guide argues AI is on balance good news for doctors, because it removes documentation burden. Does that match what you've heard? 7. Which of the five human strengths do you think you'd be naturally good at? Which would you have to work at?

The harder conversation

8. Ten to fifteen years to independent practice. Have you pictured that concretely — where you'd be at 25, at 30, at 35? 9. What's the honest financial picture, and has it actually been worked through rather than assumed? 10. Moral injury — being unable to give the care you believe a patient needs because of time or staffing. How do you think you'd cope with it? It's what makes

people leave a profession they still believe in. 11. What happens if you don't get in first time? Is there a plan, or is medicine the only plan?

The AI question, practically

12. Of the five AI-native capabilities — verify, escalate, explain, recognize the limits, contribute — which sounds most interesting? Which sounds hardest? 13. The verification skill only works on top of real clinical knowledge. Does that change how you'd approach the early, memorization-heavy years?

Testing it against reality

14. **The most valuable single action:** find a working doctor — ideally not a family friend who'll be encouraging — for an honest half-hour. Ask how they got in and whether that route still works today, what's changed about the job in the last three years, what the new doctors on their team do differently, and what they wish someone had told them before they started. 15. Can you arrange any clinical exposure — shadowing, volunteering, a hospital visit? Ideas about this job change fast on contact with the real thing.

Widening the frame

16. If medicine weren't available at all, what's the closest thing you'd choose? The answer usually reveals what you're actually drawn to. 17. Do you want *medicine*, or what medicine is made of — helping people, science, status, problem-solving under pressure? Those four lead to quite different places. Nursing, physiotherapy, paramedicine, dentistry, veterinary medicine and biomedical research all overlap with medicine but differ enormously in training length, cost and daily rhythm. 18. Which specialties interest you already? You don't need to decide, but the answer shapes which schools suit.

This is analysis and informed judgment about a fast-moving situation, not a guarantee or a prediction. Where the evidence is contested we've said so, where our own score is challenged by the data we've flagged it, and where we might be wrong we've linked to the people making the opposite case. It's a starting point for a decision — not the decision itself.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology, not archived from published editions. Each year is scored on what was known at the time. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Medicine

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned}\text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment})\end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

SURGERY — 2.1 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	3.9	4.1	4.5	1.57
Entry-path erosion	15%	1.0	1.2	1.5	0.22
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	9.0	9.0	9.0	0.1
				Total	2.1 → 2.1

EMERGENCY MEDICINE — 2.5 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.3	4.8	5.5	1.92
Entry-path erosion	15%	1.0	1.2	1.5	0.22
Physical / embodied	15%	9.0	9.0	9.0	0.15
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	9.2	9.2	9.2	0.08
				Total	2.5 → 2.5

PRIMARY CARE / FAMILY MEDICINE — 2.9 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.8	5.3	6.0	2.1
Entry-path erosion	15%	1.0	1.2	1.5	0.22
Physical / embodied	15%	8.0	8.0	8.0	0.3
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	2.9 → 2.9

ANESTHESIOLOGY — 3.1 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.6	6.1	6.6	2.31
Entry-path erosion	15%	1.0	1.2	1.5	0.22
Physical / embodied	15%	8.0	8.0	8.0	0.3
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	3.11 → 3.1

PSYCHIATRY — 3.7 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.0	6.7	7.4	2.59
Entry-path erosion	15%	1.2	1.5	1.8	0.27
Physical / embodied	15%	6.0	6.0	6.0	0.6
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	9.0	9.0	9.0	0.1
				Total	3.69 → 3.7

PATHOLOGY — 4.7 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.2	7.5	8.6	3.01
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	5.0	5.0	5.0	0.75
Human trust / accountability	15%	8.0	8.0	8.0	0.3
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	4.71 → 4.7

RADIOLOGY — 4.9 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.3	7.6	8.9	3.11
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	4.5	4.5	4.5	0.82
Human trust / accountability	15%	8.0	8.0	8.0	0.3
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	4.89 → 4.9

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Surgery

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	3.9	1.0	0.3	1.8
2025	4.1	1.2	0.3	1.9
Fall 2026	4.5	1.5	0.3	2.1

Movement decomposition: automatability contributed **+0.21** and entry-path erosion **+0.07**, for a total of **+0.3** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.